Energy MBA • Energy Markets Quantitative Analysis (EMQA)
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Assignment 1: Forecasting the Romanian day-ahead price for a bidding desk

Point and probabilistic forecasts, forecast tests and volatility — lectures 1–3

Instructor to fill in: Deadline: **TODO** • Weight in the final grade: **TODO** • Individual or teams of 2: **TODO** • Oral defence slots: **TODO** • Late policy: **TODO**. Submit on Moodle.

The situation

You work on the day-ahead desk of a Romanian energy company: a supplier, a producer or a large consumer. Every day, before 12:00 CET (Central European Time) on day $d-1$, the desk sends bids to the Romanian day-ahead market (PZU, *Piața pentru Ziua Următoare*, operated by OPCOM, the Romanian power and gas market operator) for each hour of day d. Today the desk prices its bids with "the price of yesterday" (a naive rule). Your manager asks three questions. (1) Should we replace the naive rule by a statistical model, at least in the hours that matter most for us (your *focus block*)? (2) How wide is the uncertainty we should communicate with every forecast? (3) Is the market now in a high-volatility regime that justifies a larger credit or collateral buffer? You answer with a reproducible notebook and a two-page memo that ends in a clear decision.

1 What you submit

D1 Notebook `A1_<ID>.ipynb`, built from the starter `A1_starter_EN.ipynb`. It must run with *Restart & Run All* from the course data folder, without errors, in at most 30 minutes on a normal laptop.

D2 Numbers file `my_numbers_A1_<ID>_personal.json`, written by the notebook (last cell of the starter).

D3 Memo (PDF, 11 pt, at most 2 pages): page 1 is the recommendation for your manager; page 2 is the appendix (key tables and the hand-reconciliation table). Structure in Task 8.

D4 Oral defence (about 10 minutes): the instructor picks one day and one hour of your test window and one adverse question; you explain your numbers from your own notebook.

2 Your personal parameters

Every student (or team) receives different settings. They are computed from your student ID by a fixed rule (a seed from the SHA-256 secure hash algorithm), so you always get the same values and nobody else gets the same set. Get them in either way:

- ⊡ `python make_assignment.py --id <ID> --assignment 1` (folder `repo/assignments`); team: `--id "ID1+ID2"`;
- ⊡ or the first cell of the starter notebook (`STUDENT_ID = ...`).

Parameter	Meaning
`test_start, test_end`	your 365-day test window (first day of a month between Sep 2024 and Nov 2025)
`burn_in_days`	112 days before the test window that you also forecast; their errors feed the probabilistic forecasts (Task 6)
`focus_hours`	the hours that matter for your desk (CET): night, morning ramp, solar hours or evening peak
`arx_windows`	the six calibration windows 56, 84, 112, 714, 721, 728 days (the same for everybody)
`lear_window`	your calibration window for LEAR: 364, 546 or 728 days
`error_window`	W: number of past days of errors used for HS and CP intervals: 56, 84 or 112
`garch_years`	length of the GARCH estimation sample before test_start: 3, 4 or 5 years
`check_days, check_hour`	two days and one hour you must reconcile by hand (Task 8)

Copy the printed block, unchanged, into the memo appendix. A memo with someone else's parameters receives 0 points.

3 Data, time and notation

All data are in `repo/data/clean/` (sources and download dates in `repo/data/SOURCES.md`). Use the helper `emqa_assign.py`; do not download other data and do not edit the data files.

Object	Call	Content
$P_{d,h}$	`ea.panel("RO")[0]`	day-ahead price of Romania, day d, hour h (EUR/MWh)
$L_{d,h}$	`ea.panel("RO")[1]`	actual load of Romania (mean MW in the hour)
$R_{d,h}$	`ea.panel("RO")[2]`	actual wind + solar generation of Romania (MW)
$\bar{P}_d$	`ea.daily_prices()["RO"]`	daily base price: mean of all market time units of the CET day

- ⊡ **Days and hours.** One row per delivery day d in CET (Central European Time, the time zone of the European market coupling, SDAC); columns `h00`–`h23` are CET hours, $h = 0, \ldots, 23$. Romanian local time (EET, Eastern European Time) is one hour later: `h00` is 01:00–02:00 in Bucharest.
- ⊡ **15-minute market time unit (MTU).** Since delivery day 1 October 2025 the market clears 96 quarter-hours per day. In the hourly matrix each hour after that date is the mean of its four quarter-hours. Forecast and evaluate hourly values.
- ⊡ **Information set (gate closure).** A forecast of day d is made at 12:00 CET on day $d-1$. At that time:
  - *prices*: all prices up to day $d-1$ are known (the $d-1$ auction cleared on day $d-2$);
  - *realised load L and wind+solar R*: only values already metered, i.e. up to the end of day $d-2$; day $d-1$ is *not* yet complete;
  - *fuels and carbon*: the last close before 12:00 on $d-1$, in practice the close of day $d-2$;
  - *transformations, scalers, thresholds*: fitted on these data only.

  A real desk would use the forecasts of the transmission system operator (TSO, Transelectrica); our free data contain only actual values. **Rule:** use the actual L and R of day $d-2$ (same hour) as the proxy for day d, and $d-3$, $d-7$ as further lags. `ea.leakage_check` enforces this rule.
- ⊡ **Daylight saving time (DST).** The CET day has 23 hours in March and 25 in October. The helper keeps 24 columns: the missing hour is the mean of its neighbours, the doubled hour is averaged (column `dst` marks these days).

4 Tasks

Do the tasks in order. Each task says what to compute, on which data, with which method, and what to show. The points add up to 100.

Task 1 — Know your test window [8 points]

On the 365 test days, compute the mean, standard deviation, minimum and maximum of $P_{d,h}$; the number of negative hours; the number of negative market time units (MTUs) in `ea.quarter_hours("RO")` (hourly rows before 1 October 2025, quarter-hour rows after; filter on the CET date of the index); and the share of test days with 15-minute MTU. Plot the daily base price from 1 January 2023 to the end of your test window and shade the window. In three sentences, name the market events inside your window (for example the Romania–Hungary (RO–HU) price separation of summer 2024, the dry hydro year 2025, negative prices in spring and summer, the 15-minute MTU from October 2025).

Show: table T1 (one column); figure with x-axis "date", y-axis "EUR/MWh", legend below the plot.

Task 2 — Two naive benchmarks [6 points]

For every day of burn-in + test (`fdays`) and every hour:

$$\hat{P}^{\text{naive}}_{d,h} = \begin{cases} P_{d-7,h}, & d = \text{Mon, Sat, Sun} \\ P_{d-1,h}, & d = \text{Tue–Fri} \end{cases} \qquad \hat{P}^{(7)}_{d,h} = P_{d-7,h}. \tag{1}$$

The first is the naive benchmark of the electricity-price-forecasting (EPF) literature [11, 5]; the second is the seasonal naive. Every later model is judged against $\hat{P}^{\text{naive}}$.

Show: both forecasts stored in `F` (`DataFrame`, rows `fdays`, columns `h00`..`h23`); `ea.leakage_check` passed.

Task 3 — Expert ARX model, rolling windows and window averaging [12 points]

Transform. Prices have spikes and negative values, so never take logarithms of prices. Use the variance-stabilising transformation (VST) of [10]:

$$Y_{d,h} = \operatorname{asinh}\left(\frac{P_{d,h} - m}{s}\right), \qquad \operatorname{asinh}(x) = \ln\left(x + \sqrt{x^2+1}\right), \qquad P_{d,h} = m + s\,\sinh(\hat{Y}_{d,h}), \tag{2}$$

where m is the median and s the median absolute deviation (MAD, scaled by 1.4826) of the prices in the calibration window only.

Model. For each hour h separately (24 regressions per day) estimate by ordinary least squares (OLS) the expert ARX model (autoregressive model with exogenous variables) of [13]:

$$\begin{aligned} Y_{d,h} = \beta_{h,0} + \beta_{h,1}Y_{d-1,h} + \beta_{h,2}Y_{d-2,h} + \beta_{h,3}Y_{d-7,h} + \beta_{h,4}Y_{d-1,23} + \beta_{h,5}Y^{\min}_{d-1} + \beta_{h,6}Y^{\max}_{d-1} \\ + \beta_{h,7}\tilde{L}_{d,h} + \beta_{h,8}\tilde{R}_{d,h} + \beta_{h,9}D^{\text{Mon}}_{d} + \beta_{h,10}D^{\text{Sat}}_{d} + \beta_{h,11}D^{\text{Sun}}_{d} + \varepsilon_{d,h} \end{aligned} \tag{3}$$

where $Y_{d-1,23}$ is the last hour of the previous day; $Y^{\min}_{d-1}, Y^{\max}_{d-1}$ are the minimum and maximum over the 24 hours of day $d-1$; $\tilde{L}_{d,h} = L_{d-2,h}$ and $\tilde{R}_{d,h} = R_{d-2,h}$ are the proxies (the last metered day; standardised with the mean and standard deviation of the calibration window, rows up to $d-2$); D are 0/1 dummies for Monday, Saturday and Sunday; $\varepsilon_{d,h}$ is the error.

Rolling calibration. For each forecast day d and each window length $T \in \{56, 84, 112, 714, 721, 728\}$, estimate on the T target days $d-T, \ldots, d-1$, then forecast day d; move one day forward and re-estimate (daily recalibration). **Averaging.** The combined forecast is the simple mean over the six windows, $\hat{P}^{\text{AW}}_{d,h} = \frac{1}{6}\sum_T \hat{P}^{(T)}_{d,h}$ [3].

Show: seven forecasts in `F`: ARX(56), …, ARX(728) and AW-ARX; `ea.leakage_check` passed for ARX(56).

Task 4 — LEAR, the strong benchmark [10 points]

LEAR (LASSO-estimated autoregression; LASSO = least absolute shrinkage and selection operator) [9, 5] uses many candidate regressors and lets the LASSO penalty switch off the useless ones. For each hour h:

$$\hat{\boldsymbol{\beta}}_h = \arg\min_{\boldsymbol{\beta}} \Big\{ \sum_{d \in \text{cal}} \big(Y_{d,h} - \mathbf{x}_d^\top \boldsymbol{\beta}\big)^2 + \lambda_h \sum_j |\beta_j| \Big\}, \tag{4}$$

where $\mathbf{x}_d$ has 247 entries: $Y_{d-k,0}, \ldots, Y_{d-k,23}$ for $k = 1, 2, 3, 7$ (96); load and wind+solar proxies of $d-2, d-3, d-7$ for all 24 hours, after the same asinh transform (144); 7 day-of-week dummies. All columns are standardised with calibration-window means and standard deviations. $\lambda_h \geq 0$ is chosen by the Akaike information criterion (AIC): `LassoLarsIC(criterion="aic")` in `scikit-learn`. Use your `lear_window`; recalibrate daily (weekly recalibration is allowed if you say so and report the run time).

Show: LEAR forecasts in `F`; the number of non-zero coefficients for hour `check_hour` on your first test day.

Task 5 — Accuracy and Diebold–Mariano tests [16 points]

On the test window only (never on the burn-in), with errors $e_{d,h} = P_{d,h} - \hat{P}_{d,h}$ and N the number of (day, hour) pairs:

$$\text{MAE} = \frac{1}{N}\sum |e_{d,h}|, \qquad \text{RMSE} = \sqrt{\frac{1}{N}\sum e_{d,h}^2}, \qquad \text{rMAE} = \frac{\text{MAE}_{\text{model}}}{\text{MAE}_{\text{naive}}}. \tag{5}$$

MAE is the mean absolute error, RMSE the root mean squared error, rMAE the relative MAE [5] (below 1: better than naive). Do not use the mean absolute percentage error (MAPE): it explodes when prices are near zero. Report all models: (a) all 24 hours, (b) your focus block, (c) each calendar year inside your window, (d) each quarter for Naive, AW-ARX and LEAR.

Diebold–Mariano (DM) test [1], multivariate version [13]. For models A and B and each test day d:

$$\Delta_d = \frac{1}{|H|}\sum_{h \in H} |e^A_{d,h}| - \frac{1}{|H|}\sum_{h \in H} |e^B_{d,h}|, \qquad \text{DM} = \frac{\bar{\Delta}}{\sqrt{\hat{\sigma}^2_{\text{NW}}/n}}, \qquad p = 1 - \Phi(\text{DM}), \tag{6}$$

where H is the set of hours (all 24, or your focus block), n the number of test days, $\bar{\Delta}$ the mean of Δ_d, $\hat{\sigma}^2_{\text{NW}}$ the Newey–West (heteroskedasticity- and autocorrelation-consistent, HAC) variance of Δ_d with 7 lags, and Φ the standard normal distribution function. A small p (e.g. below 0.05) means: B is significantly more accurate than A. Show two 6×6 matrices of p-values (all hours; focus block) for Naive, seasonal naive, ARX(56), ARX(728), AW-ARX, LEAR.

Show: table T5 (rows = models; columns MAE, RMSE, rMAE overall, focus block, per year); quarterly MAE table; two DM matrices; two sentences: which model wins, and is the win significant?

Task 6 — Probabilistic forecasts: historical simulation and conformal prediction [14 points]

Turn a point forecast into 99 quantiles $\hat{q}^\tau_{d,h}$, $\tau = 0.01, \ldots, 0.99$, using the errors of the same hour on the previous W days (`error_window`), $e_{d-W,h}, \ldots, e_{d-1,h}$ [7, 6]:

$$\text{HS:} \quad \hat{q}^\tau_{d,h} = \hat{P}_{d,h} + Q_\tau(e_{d-W,h}, \ldots, e_{d-1,h}), \tag{7}$$

$$\text{CP:} \quad \hat{q}^\tau_{d,h} = \hat{P}_{d,h} + \text{sign}(\tau - 0.5)\, Q_{|2\tau-1|}(|e_{d-W,h}|, \ldots, |e_{d-1,h}|), \tag{8}$$

where $Q_a(\cdot)$ is the empirical a-quantile (`numpy.quantile`, default linear interpolation). HS (historical simulation) uses the signed errors, so the interval can be asymmetric; CP (conformal prediction) uses absolute errors, so it is symmetric around $\hat{P}_{d,h}$. Apply HS and CP to Naive, AW-ARX and LEAR (6 combinations). **Pinball score** for quantile level τ and actual price y, and its average, the average

pinball score (APS):

$$\mathrm{PS}(\hat{q}^\tau, y, \tau) = (\mathbf{1}\{y < \hat{q}^\tau\} - \tau)(\hat{q}^\tau - y), \qquad \mathrm{APS} = \frac{1}{99\,N} \sum_{d,h} \sum_{\tau} \mathrm{PS}(\hat{q}^\tau_{d,h}, P_{d,h}, \tau), \qquad \mathrm{CRPS} \approx 2 \cdot \mathrm{APS}, \tag{9}$$

where $\mathbf{1}\{\cdot\}$ is 1 if the condition holds and 0 otherwise; CRPS is the continuous ranked probability score [2]. Lower is better. **Coverage**: share of (day, hour) pairs with $P_{d,h}$ inside the 50% interval $[\hat{q}^{0.25}, \hat{q}^{0.75}]$ and the 90% interval $[\hat{q}^{0.05}, \hat{q}^{0.95}]$. **Kupiec test** [4] on the focus block for the 90% interval, with x misses out of n cases and nominal miss rate $p_0 = 0.10$:

$$\mathrm{LR}_{\mathrm{UC}} = -2 \ln \frac{(1-p_0)^{n-x} p_0^x}{(1-\hat{p})^{n-x} \hat{p}^x}, \qquad \hat{p} = x/n, \qquad \mathrm{LR}_{\mathrm{UC}} \sim \chi^2_1 \text{ under } H_0. \tag{10}$$

Show: table T6 (rows = 6 combinations; columns APS, APS on focus block, coverage 50%, coverage 90%, coverage 90% on focus block, Kupiec p-value, mean width of the 90% interval on the focus block); one figure: one test week, actual price and the 50% and 90% bands of your best combination.

Task 7 — Volatility: GARCH, GJR-GARCH, EGARCH [12 points]

GARCH is the generalised autoregressive conditional heteroskedasticity model; GJR-GARCH its asymmetric version of Glosten, Jagannathan and Runkle; EGARCH the exponential GARCH. Use daily base prices. Because prices can be zero or negative, do not use log-returns; use

$$r_t = 100\,[\,\mathrm{asinh}(\bar{P}_t) - \mathrm{asinh}(\bar{P}_{t-1})], \qquad \varepsilon_t = r_t - \mu - \delta_{\mathrm{dow}(t)} = \sigma_t z_t, \quad z_t \sim t_\nu \text{ (standardised)}, \tag{11}$$

where $\delta_{\mathrm{dow}(t)}$ is the mean of r on that weekday, computed on the estimation sample only (subtract it before fitting), σ_t the conditional volatility and ν the degrees of freedom. Estimate by maximum likelihood (package `arch`) on the `garch_years` before `test_start`:

$$\text{GARCH(1,1):} \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \tag{12}$$

$$\text{GJR(1,1,1):} \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 \mathbf{1}\{\varepsilon_{t-1} < 0\} + \beta \sigma_{t-1}^2, \tag{13}$$

$$\text{EGARCH(1,1,1):} \quad \ln \sigma_t^2 = \omega + \alpha(|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2. \tag{14}$$

In GJR, $\gamma > 0$ means falls raise volatility more ("leverage", usual for stocks); $\gamma < 0$ means price jumps raise volatility more ("inverse leverage", often reported for gas and power). In EGARCH the signs are reversed in meaning: $\gamma < 0$ = falls raise volatility more. **Test, do not assume**: report γ and its p-value. Persistence is $\alpha + \beta + \gamma/2$ (GJR), $\alpha + \beta$ (GARCH), β (EGARCH); half-life $= \ln 0.5 / \ln(\text{persistence})$ days. Then, with the parameters fixed, filter through the test window and compare the one-day-ahead variance forecasts $\hat{\sigma}_t^2$ with the proxy ε_t^2 by the quasi-likelihood loss QLIKE $= \frac{1}{n}\sum_t (\ln \hat{\sigma}_t^2 + \varepsilon_t^2/\hat{\sigma}_t^2)$ (lower is better) [8]. Finally translate into money the one-day-ahead forecast $\hat{\sigma}_T$ for the last test day T: the approximate 99% one-day upward move of the base price, $\bar{P}_{T-1}\,[\exp(z_{0.99}\hat{\sigma}_T/100) - 1]$, with $\bar{P}_{T-1}$ the actual base price of the day before and $z_{0.99} = t_\nu^{-1}(0.99)\sqrt{(\nu-2)/\nu}$ the 99% quantile of the standardised t_ν.

Show: table T7 (rows = 3 models; columns $\omega, \alpha, \gamma, \beta, \nu$, log-likelihood, AIC, BIC (Bayesian information criterion), persistence, half-life, p-value of γ, QLIKE on test); figure of $|\varepsilon_t|$ and the three $\hat{\sigma}_t$ on the test window; the 99% move in EUR/MWh.

Task 8 — The memo: decision, reconciliation, your company [22 points]

Page 1 (recommendation, at most 450 words):

(a) **Decision** in the first two lines: which model the desk should use for the focus block, from which date, and what it replaces.

(b) **Evidence** (three bullets, each with a number from your notebook): accuracy gain vs naive (rMAE, EUR/MWh), DM p-value, the 90% interval width and its coverage.

(c) **Risk**: what the GARCH result says about the current regime, and your buffer recommendation in EUR/MWh.

(d) **Limits**: the load/renewables proxy, one period where your model failed, and what you would add next.

(e) **How this applies in my company** (at most 120 words). Public information only: never use or disclose confidential data. Example: "our supplier buys about X% of its portfolio on PZU in the evening peak" is acceptable only if X is public; otherwise describe the mechanism without numbers.

Page 2 (appendix): your parameter block; excerpts of T5, T6, T7; and the **hand-reconciliation table**: for each of your two `check_days` at `check_hour`, write by hand the actual price, the Naive, AW-ARX and LEAR forecasts, each absolute error, the LEAR+CP 5% and 95% quantiles, whether the price was inside, and the pinball score at $\tau = 0.95$, with the arithmetic shown (e.g. $(0-0.95)(278.9-394.9) = 110.2$).

5 Exact conventions (answers to the questions students usually ask)

C1 **Names in `F`** (a Python `dict` of `DataFrame`s, rows `fdays`, columns `h00`..`h23`): `"Naive (Weron)"`, `"Seasonal naive d-7"`, `"ARX(56)"`, ..., `"ARX(728)"`, `"AW-ARX(6 windows)"`, `"LEAR(<lear_window>)"`. Columns of T5 per year: `MAE_2025`, `RMSE_2025`, `rMAE_2025`.

C2 **VST**: m and s are computed once per forecast day and window, *pooled over all 24 hours* of the price rows $d-T-7, \ldots, d-1$ (targets plus lag rows; hence 63 days in the worked example for $T = 56$), and used for every hour. **ARX exogenous**: L and R are z-scored with one mean and one standard deviation pooled over the same rows and hours, but only up to row $d-2$ (no asinh). **LEAR exogenous**: L and R get their own asinh transform (own median/MAD, same rows up to $d-2$), then every one of the 247 columns, dummies included, is z-scored with calibration means and standard deviations. For $h = 23$ the ARX regressor $Y_{d-1,23}$ appears twice: drop the duplicate (11 regressors).

C3 **Years and quarters**: calendar years and calendar quarters intersected with your window (partial periods allowed; report the number of days). rMAE always divides by the Naive MAE on the same subset (same days, same hours).

C4 **DM matrix**: the cell in row A, column B holds $p = 1 - \Phi(\mathrm{DM})$ with $\Delta_d = \text{mean } |e^A| - \text{mean } |e^B|$; the diagonal is empty.

C5 **Kupiec on the focus block**: n = test days $\times$ number of focus hours; x = (day, hour) pairs outside $[\hat{q}^{0.05}, \hat{q}^{0.95}]$ on either side; "inside" includes the bounds; if $x = 0$ use $0 \cdot \ln 0 = 0$. The pairs within a day are not independent: mention it. **Best combination** (for the figure) = lowest APS on the test window; any test week. Report CRPS $\approx 2 \cdot$APS as an extra column.

C6 **GARCH in `arch`**: `garch_years` are calendar years (`test_start - pd.DateOffset(years=g)`). Subtract the estimation-sample weekday means from all r_t (estimation and test). Fit `arch_model(r, mean="Constant", vol="GARCH", p=1, o=0 or 1, q=1, dist="t")` on the series up to `test_end` with `.fit(last_obs=test_start)` (parameters use only the estimation sample), then `.forecast(horizon=1, start=<last estimation day>, reindex=False)`: the forecast made at day t is the variance of day $t+1$. For EGARCH use `vol="EGARCH"`. The p-value of γ is "n/a" for GARCH(1,1).

C7 **Leakage check**: run `ea.leakage_check` at least for Naive, ARX(56) and LEAR on the first test day. **Where to work**: keep your notebook in `repo/assignments/A1/` next to the starter (it imports `emqa_assign` from the parent folder). `MODE="latest"`: only a clean run is required; tables may contain empty cells (e.g. one quarter only). **Hand reconciliation**: typed arithmetic is fine; the pinball score is that of the LEAR+CP quantile $\hat{q}^{0.95}$.

6 Worked example (real data): the rolling window and an HS/CP interval

Forecast day d = Tuesday 4 March 2025, hour `h19` (19:00–20:00 CET), information cut-off 3 March 2025, 12:00 CET.

(1) Calibration window for ARX(56).

Item	Dates	Why
target days (56)	2025-01-07 … 2025-03-03	$d-56,\dots,d-1$
oldest price row used	2024-12-31	lag 7 of the first target
m, s of the VST	prices 2024-12-31 … 2025-03-03	never the test day
L, R (proxies, scaling)	rows up to 2025-03-02	$d-1$ not yet metered
next step	targets 2025-01-08 … 2025-03-04	forecast 5 March (roll one day)

(2) Regressors of d for hour h19 (before the VST), from the course data.

$P_{d-1,19}$	$P_{d-2,19}$	$P_{d-7,19}$	$P_{d-1,23}$	$P^{\min}_{d-1}$	$P^{\max}_{d-1}$	$L_{d-2,19}$	$R_{d-2,19}$	$D^{\text{Mon}}, D^{\text{Sat}}, D^{\text{Sun}}$	actual $P_{d,19}$
198.18	178.86	438.63	151.23	46.34	286.21	7 119 MW	843 MW	0/0/0	231.65

The naive forecast is $P_{d-1,19} = 198.18$ (Tuesday uses $d-1$), error $231.65 - 198.18 = 33.47$ EUR/MWh.

(3) An 80% interval from the last $W = 10$ naive errors at h19 (22 Feb – 3 Mar 2025; $W = 10$ only to keep the example short):

sorted errors	−246.5	−159.4	−66.0	−16.8	−13.3	55.1	64.3	66.0	81.1	140.7
sorted $\lvert e\rvert$	13.3	16.8	55.1	64.3	66.0	66.0	81.1	140.7	159.4	246.5

HS: $Q_{0.10}(e) = -168.1$ and $Q_{0.90}(e) = 87.0$ (linear interpolation), so the interval is $[198.2-168.1,\ 198.2+87.0] = [30.1,\ 285.2]$. CP: $Q_{0.80}(|e|) = 144.4$, so $[198.2 - 144.4,\ 198.2 + 144.4] = [53.7,\ 342.6]$. The actual 231.65 lies inside both. Pinball at $\tau = 0.90$ for HS: $\hat{q} = 285.2 > y = 231.65$, so $\text{PS} = (1 - 0.9)(285.2 - 231.65) = 5.36$.

(4) DM in one line. If on a test day the mean $|e|$ over your focus hours is 30 for Naive (A) and 22 for LEAR (B), then $\Delta_d = +8$: LEAR was better that day. The test asks whether the mean of these daily differences is significantly above zero.

7 Rules: reproducibility and use of AI tools

R1 Runs end-to-end: *Restart & Run All* works on the course data with the packages of the course environment (`numpy`, `pandas`, `scikit-learn`, `statsmodels`, `arch`, `matplotlib`).

R2 No hard-coded dates or numbers: dates come from `p`. The instructor re-runs every notebook with `MODE = "latest"` on data downloaded after the deadline (an unseen hold-out); it must run.

R3 Data version: the notebook prints `ea.data_version()` (file hashes).

R4 No look-ahead: `ea.leakage_check` passes for every model you report.

R5 Every number in the memo is in `MY_NUMBERS`. A number that is not, or that differs from the notebook, costs 2 points (up to 10).

R6 AI tools (ChatGPT, Claude, Copilot, …) are allowed for code, with disclosure: add a cell "AI use" that names the tool and what it did. They are not allowed to write the memo's numbers or conclusions. You must be able to explain every line at the oral defence.

If the notebook does not run (R1), Tasks 1–7 are capped at 50%. If you cannot explain a result at the oral defence, the instructor may set the points of that task to 0.

8 Common pitfalls

- ⊡ **Look-ahead leakage**: using $L_{d,h}$ or $R_{d,h}$ of the forecast day; fitting the VST, the scaler or the LASSO on data that include day d; choosing a model on the test window and then reporting its test accuracy as if it were out-of-sample.
- ⊡ **Logarithms of prices** or log-returns: RO had 196 negative hours in 2025 alone. Use asinh.
- ⊡ **MAPE** with prices near zero; deleting negative prices to make the errors look smaller.
- ⊡ **Mixing 15-minute and hourly data**, or UTC and CET hours (a one-hour shift destroys the hourly models).
- ⊡ **In-sample fit reported as forecast accuracy**; R^2 instead of MAE/RMSE on the test window.
- ⊡ **Reading DM backwards**: our p-value is for "column better than row".
- ⊡ **Generic text**: "electricity prices are volatile" earns nothing. Every claim needs a number from your window.

9 Before you submit: checklist

☐ Parameter block printed in the notebook and copied into the memo appendix.

☐ *Restart & Run All* works; run time noted; `MODE = "latest"` also runs.

☐ `ea.data_version()` printed; `ea.leakage_check` passed for Naive, ARX(56), LEAR.

☐ Tables T1, T5, T6, T7, two DM matrices, three figures (axes with units, legend below).

☐ Memo $\leq$ 2 pages; decision in the first two lines; every number traceable to `MY_NUMBERS`.

☐ Hand-reconciliation table with arithmetic; "AI use" cell; no confidential company data.

10 Frequently asked questions

Which time zone?
CET/CEST everywhere (the files are already converted). Bucharest time = CET + 1 h. Do not convert again.

Can I forecast the 96 quarter-hours?
Not required. The models and the grading use the hourly matrix. A 15-minute analysis is welcome as an extra figure but earns no extra points.

What about negative prices and the price floor?
Keep them; they are real. The harmonised minimum clearing price of the day-ahead coupling (SDAC) is −500 EUR/MWh and became −600 EUR/MWh for delivery from 29 May 2026 (OPCOM notice of 7 May 2026).

Are there missing days?
Not in `ea.panel()` from 2022 on (checked). If your code finds a gap, you probably mixed time zones. Days before 2019 have gaps; you do not need them.

Why the burn-in?
HS and CP need W past forecast errors. Without the 112 burn-in days your first test days would have no interval. Accuracy is measured on the test window only.

LEAR is slow. What can I do?
About 1 minute on 8 cores with `joblib.Parallel` over days; 10–20 minutes on one core. Weekly recalibration is allowed if declared.

Can I add XGBoost, Prophet or a neural network?
Yes, in addition to (not instead of) the required models, with the same rolling scheme. No extra points in Assignment 1; machine learning is graded in Assignment 2.

My DM p-value is 1.000. Is that an error?
No. It means the column model is clearly *worse* than the row model; read the symmetric cell.

Which quantile convention?
`numpy.quantile` default (linear). Small differences from other conventions are not penalised.

Why asinh for GARCH and not percentage returns?
Percentage changes explode when $\bar{P}_{t-1}$ is near zero. For prices well above 1 EUR/MWh, $\operatorname{asinh}(P) \approx \ln(2P)$, so r_t behaves like a log-return.

My GJR γ is positive, but the lecture said "inverse leverage".
Report what your data say and discuss why (e.g. daily base prices after weekday adjustment vs. hourly spikes). There is no "right sign" for the grade.

Can I use R or Excel?
The graded notebook must be Python (TODO: instructor may allow R).

Team work?
If allowed (see the box on page 1), use `--id "ID1+ID2"`; both members defend orally and must explain any part.

11 Grading rubric (100 points)

Task	Pts	Excellent (90–100%)	Adequate (50–70%)	Poor (0–30%)
1 Test window	8	T1 complete incl. negative MTUs; clean figure; events named and dated from the data	table complete, events generic	wrong window or missing table
2 Naive	6	both rules exact (weekday logic), leakage check passed	one rule wrong on some weekdays	missing or uses day d
3 ARX + averaging	12	exact regressors, VST fitted in-window, 6 rolling windows, daily recalibration, AW	fixed window or some regressors missing	in-sample fit only, or leakage
4 LEAR	10	247 regressors, AIC-selected λ per hour, rolling, sparsity reported	fewer regressors or fixed λ	not rolling / not run
5 Accuracy, DM	16	all metrics by year, quarter and focus block; HAC DM both matrices; correct reading; significance discussed	metrics right, DM missing one matrix or misread once	metrics on burn-in, MAPE, no DM
6 Probabilistic	14	6 combinations, correct HS/CP, APS, coverage, Kupiec, figure; calibration vs sharpness discussed	HS or CP only, no Kupiec	intervals from in-sample residuals or future errors
7 Volatility	12	3 models with t errors, γ tested and interpreted, persistence, QLIKE out-of-sample, 99% move in EUR	in-sample only or no money translation	log-returns of prices; no estimation window
8a Decision	10	clear decision first; 3 numbers; limits and failure period honest	decision vague or numbers without meaning	generic text, no decision
8b Reconciliation	6	both days by hand, arithmetic shown, matches notebook	one day or arithmetic missing	missing or inconsistent
8c My company	6	concrete mechanism in the student's job, public data only	plausible but generic	absent, or confidential data

Penalties and gates: R1 (does not run) caps Tasks 1–7 at 50%; R5 costs 2 points per untraceable number (up to 10); wrong personal parameters = 0. A short oral defence confirms authorship and may reduce, never increase, the points of a task.

12 Further reading (papers of the course instructor and co-authors)

- A.-V. Andrei, G. Velev, F.-M. Toma, D.T. Pele, S. Lessmann (2024). *Energy Price Modelling: A Comparative Evaluation of four Generations of Forecasting Methods*, arXiv:2411.03372 — Tasks 2–5.
- *Explainable AI-Based Early Warning System for Electricity Price Spikes: Evidence from the Romanian Market*, PICBE 2026, doi:10.2478/picbe-2026-0352 — Tasks 1 and 7.

⊡ *Evaluating Intervention Effectiveness during Extreme Price Events: Romania's 2024 Energy Crisis*, PICBE 2026, doi:10.2478/picbe-2026-0013 — Task 1.

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